A PROJECT REPORT

On

Create a platform for users to buy and sell used electric vehicle parts, promoting recycling

Submitted in partial fulfillment of the requirement of
University of Mumbai for

Web Programming Mini Project
In
Computer Engineering (IV SEM)

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This project entitled "Create a platform for users to buy and sell used electric vehicle parts, promoting recycling" by Kshitij Kiran Koli, Nawfal Zameer Deshmukh, Jais Jacob Sunny and Jijo George are approved for the degree of Bachelor of Engineering in Computer Engineering.

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DECLARATION

We declare that this written submission for the Web Programming Mini Project entitled "Create a platform for users to buy and sell used electric vehicle parts, promoting recycling" represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission have not been taken when needed.

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Abstract

The **Voltmart system** is a web-based application designed to provide a dedicated and efficient platform for buying and selling used electric vehicle (EV) parts. With the rapid advancement of electric vehicle technology and the increasing global adoption of EVs, there has been a significant rise in the demand for spare parts and maintenance services. However, purchasing new EV components is often costly, creating a financial burden for users. Simultaneously, many used EV parts that are still functional are discarded due to the absence of a proper resale mechanism, leading to unnecessary waste and negative environmental impact.

The proposed system addresses these challenges by introducing an online marketplace where users can list their used EV parts and buyers can browse, compare, and purchase them at affordable prices. The platform allows users to create accounts, securely log in, and manage their product listings with ease. It also provides efficient search and filtering functionality, enabling users to quickly find specific components based on their requirements.

The system is developed using **HyperText Markup Language (HTML)**, **Cascading Style Sheets (CSS)**, and **JavaScript** to build an interactive, responsive, and user-friendly frontend interface. The backend is implemented using **Node.js**, which handles server-side logic, API integration, and user authentication. For database management and real-time data handling, **Firebase** is used, offering scalable cloud storage, secure authentication services, and efficient data synchronization.

Unlike traditional methods where buying and selling used parts is often unorganized and unreliable, Voltmart provides a structured, centralized, and secure solution. It promotes the reuse and recycling of EV components, thereby reducing electronic waste and supporting environmentally sustainable practices. Additionally, the platform makes EV maintenance more economical and accessible to a broader range of users.

Overall, the Voltmart system demonstrates how modern web technologies and cloud-based solutions can be effectively utilized to solve real-world problems while enhancing cost efficiency, accessibility, and environmental sustainability.

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Chapter 1

Introduction to Voltmart System

1.1 Introduction

In recent years, **electric vehicles (EVs)** have gained significant popularity due to their eco-friendly nature and reduced dependency on fossil fuels. Governments and organizations across the world are actively promoting EV adoption to reduce carbon emissions and combat climate change. As a result, the number of EV users has increased rapidly, leading to a transformation in the transportation sector.

With this rapid growth, the demand for maintenance services and spare parts has also increased. However, EV components such as batteries, controllers, motors, and sensors are often expensive, making maintenance costly for users. Many EV owners face difficulties in repairing or replacing parts due to high prices and limited availability of affordable alternatives.

At the same time, a large number of EV parts that are still functional are discarded due to minor faults, upgrades, or lack of awareness about reuse. These components can often be repaired or reused, but due to the absence of a proper and organized resale platform, they go to waste. This not only results in environmental pollution but also leads to the loss of valuable resources.

To address these challenges, the **Voltmart system** is developed as a web-based application that connects buyers and sellers of used EV parts through a centralized digital platform. It acts as an online marketplace where users can easily list their products, explore available items, and make cost-effective purchases.

The system is built using **HyperText Markup Language (HTML)**, **Cascading Style Sheets (CSS)**, and **JavaScript (JS)** for designing a responsive, interactive, and user-friendly frontend interface. The backend is developed using **Node.js**, which handles server-side operations, request processing, and application logic. For database management and real-time data storage, **Firebase (a cloud-based Backend-as-a-Service platform by Google)** is used, providing features such as secure authentication, real-time database updates, and scalable cloud storage.

The platform ensures ease of use by offering a simple and intuitive interface, a secure login and authentication system, and efficient data handling. Users can upload product details, search for specific EV parts, and manage their listings effectively.

By promoting reuse and recycling of EV components, the Voltmart system contributes to environmental sustainability and helps reduce electronic waste. Additionally, it provides users with affordable alternatives to expensive new parts, making EV maintenance more economical and accessible.

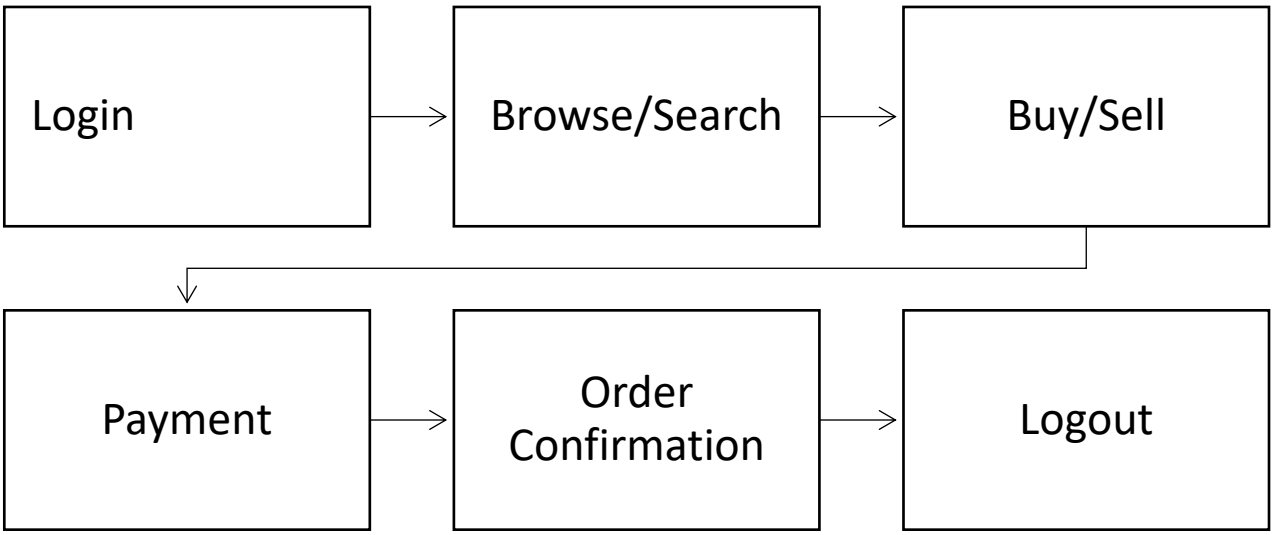


Fig 1.1: Voltmart Workflow

The Figure 1.1 represents the basic workflow of the **Voltmart**. The workflow of the system begins when a user opens the platform and either registers or logs in, after which their credentials are verified through the database. Once authenticated, the user accesses the dashboard where they can browse available electric vehicle parts, search for specific items, or choose to sell a product by entering its details, which are then stored in the database and made visible to others. Buyers can view product details, add items to the cart, and proceed to checkout, where payment is processed securely; upon successful payment, the order is confirmed and recorded in the system. Throughout this process, the backend manages user data, product listings, and transactions, ensuring smooth operation until the user logs out, ending the session.

Chapter 2

Objective

2.1 Problem Definition

The increasing adoption of electric vehicles has created a growing demand for spare parts and maintenance solutions. However, there is a lack of a dedicated platform for buying and selling used EV components. This results in several challenges.

Firstly, many usable EV parts are discarded due to the absence of a resale system. This leads to wastage of valuable resources and contributes to environmental pollution.

Secondly, users are forced to purchase new parts at higher costs, even when used alternatives are available. This increases the overall cost of EV maintenance and reduces affordability.

Thirdly, existing platforms are not specifically designed for EV parts, making it difficult for users to find relevant components. There is no proper categorization, search functionality, or trust mechanism.

To overcome these issues, the Voltmart system is proposed. It provides a centralized, user-friendly platform where users can easily buy and sell used EV parts. The system improves accessibility, reduces cost, and promotes sustainability.

2.2. System Requirements

Hardware Requirements:

1. Processor: Intel Core i3 or higher
2. RAM: Minimum 4 GB
3. Hard Disk: 500 MB free space
4. Display: 1366 × 768 resolution

Software Requirements:

1. Operating System: Windows / Linux / MacOS
2. Frontend: HTML, CSS, JavaScript
3. Backend: Node.js
4. Database: NoSQL
5. Server: XAMPP
6. Browser: Google Chrome/ Microsoft Edge

Chapter 3

Methodology

3.1 Database schema and ER diagram

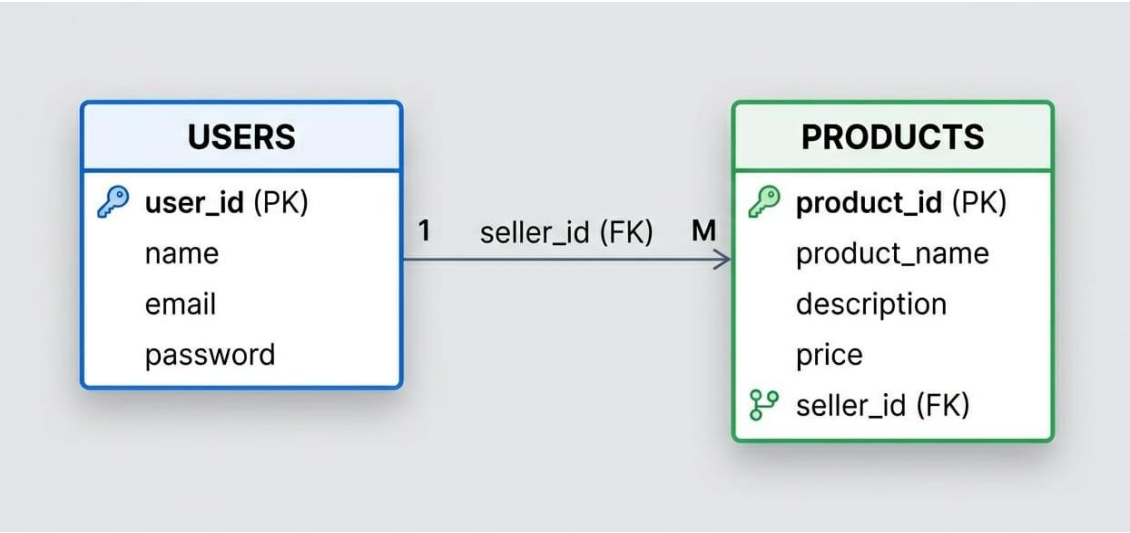
The Voltmart system uses a relational database to store and manage user and product data efficiently. The database is implemented using MySQL, which provides reliable and scalable data management.
The system consists of two primary entities:

1. Users Entity:

This table stores information about registered users. It includes fields such as user ID, name, email, and password. Each user is uniquely identified by a user ID.

2. Products Entity:

This table stores details of EV parts listed for sale. It includes product ID, product name, description, price, and seller ID.
The relationship between Users and Products is one-to-many, meaning a single user can list multiple products, but each product belongs to only one user.
This database structure ensures efficient storage, quick retrieval, and proper management of data. It also supports CRUD operations such as creating, reading, updating, and deleting records.



3.1 ER Diagram of Voltmart

The ER diagram represents the database structure of an e-commerce platform called Voltmart, designed to manage users and the products they sell in a clear and organized way. It consists of two main entities: USERS and PRODUCTS, each with its own set of attributes and a defined relationship between them. The USERS entity stores information about individuals registered on Voltmart, including a unique user_id which serves as the primary key to identify each user distinctly. Along with this, it contains attributes such as name, email, and password, which are essential for user identification, communication, and authentication within the platform.

Every user in Voltmart must have a unique ID, ensuring there is no duplication and allowing the system to manage accounts efficiently.

The PRODUCTS entity, on the other hand, stores all the items available for sale on Voltmart. Each product is uniquely identified by a product_id, which acts as the primary key for the table. Additional attributes include product_name, description, and price, which provide detailed information about each item listed on the platform. A crucial attribute in this entity is seller_id, which is a foreign key that references the user_id in the USERS table. This connection establishes a relationship between the two entities, linking each product to the user who is selling it. In Voltmart, this ensures that every product listed has a valid seller associated with it, preventing orphan records and maintaining data consistency.

The relationship between USERS and PRODUCTS is defined as one-to-many (1:M). This means that a single user on Voltmart can act as a seller and list multiple products for sale, while each individual product can only be associated with one seller. For example, if a user named Alice registers on Voltmart and receives a user_id of 101, she can list several products such as a phone, laptop, or accessories, all of which will have seller_id = 101 in the PRODUCTS table. This relationship is fundamental to how the platform operates, as it reflects the real-world behavior of sellers managing multiple listings under one account.

Overall, this ER diagram provides a foundational structure for Voltmart's database by ensuring data integrity, enforcing relationships between users and their products, and supporting scalability as the platform grows. It allows Voltmart to efficiently manage user accounts and product listings while maintaining accurate and reliable connections between them.

3.2 Algorithms or Technologies Used and their description

This section describes the implementation requirements of the proposed system, including the algorithms and technologies used for developing the platform. The "Voltmart" system is designed using a combination of front-end, back-end, and database technologies to ensure efficient performance and user-friendly interaction.

The implementation involves the use of web development technologies to create an interactive interface, server-side scripting to handle application logic, and database management systems to store and retrieve data. These technologies work together to provide functionalities such as user authentication, product listing, searching, and data management.

The following subsections provide a detailed description of the algorithms and technologies used in the system.

3.2.1 Algorithm / Tools/ Techniques

The Voltmart system uses modern web technologies to ensure smooth functionality and user experience.

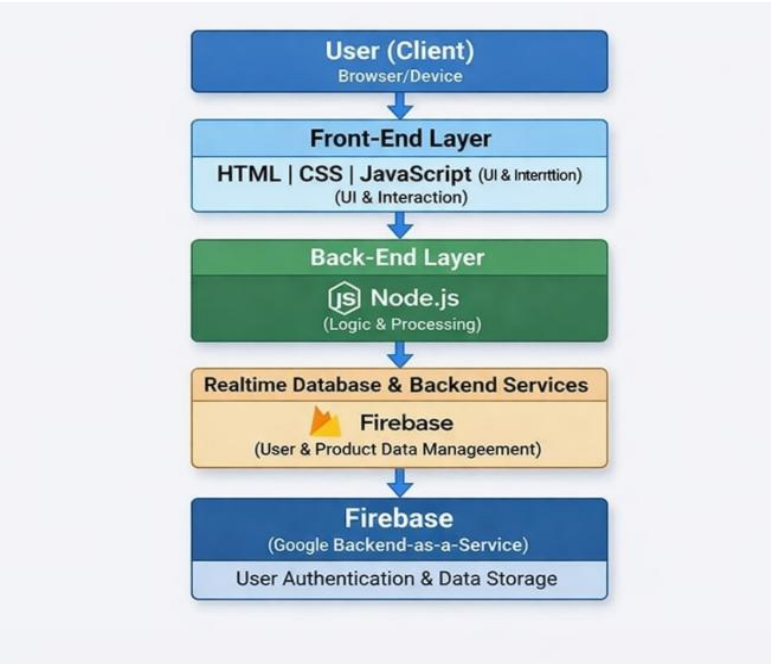
- **HTML (HyperText Markup Language):** Used to create the structure of web pages. It defines elements such as forms, buttons, and layouts.

- **CSS (Cascading Style Sheets):** Used to design the user interface. It enhances the visual appearance by adding colors, layouts, and responsiveness.

- **JavaScript:** Provides interactivity to the system. It is used for form validation, dynamic content updates, and user interaction.

- **Node.js:** Used as the backend language. It handles server-side processing, user authentication, and database communication.

- **NOSQL:** Used for storing and managing data. It ensures efficient handling of user and product information.



3.2.2 Block Diagram

The block diagram of the Voltmart system illustrates the overall architecture of the web application. At the top, users interact with the system through a browser or device. The **Front-End layer**, built using **HyperText Markup Language (HTML)**, **Cascading Style Sheets (CSS)**, and **JavaScript (JS)**, provides a responsive and interactive user interface. User requests are then sent to the **Back-End layer**, developed using **Node.js**, which handles application logic, request processing, and communication between components. The backend further connects to **Firebase**, a cloud-based **Backend-as-a-Service (BaaS)** platform.

Firebase is responsible for managing **real-time database operations**, ensuring that user and product data is stored and updated instantly. It also provides **secure authentication services** for user login and registration, along with **cloud storage** for storing images and related data. The entire system follows a structured flow where data moves from the user interface to the backend, gets processed and stored in Firebase, and then the response is sent back to the user. This architecture ensures high performance, scalability, security, and efficient data management for the application.

3.2.3 Hardware and Software Specifications

The experiment setup is carried out on a computer system which has the different hardware and software specifications as given in Table 3.2 and Table 3.3 respectively.

Table 3.2 Hardware details

Processor	Intel i3 / AMD Ryzen3 or better
Storage	500 GB
RAM	4 GB

Table 3.3 Software details

Operating System	Windows 10/11, Linux or MacOS
Tools	VS Code, XAMPP
Browser	Chrome
Database	MySQL

Chapter 4
Implementation

4.1 System Overview

- **Frontend**
  The frontend of the Voltmart system is developed using HyperText Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript (JS). It provides a responsive and user-friendly interface that allows users to interact with the platform easily. Users can register, log in, browse available EV parts, search for specific components, and list their own products for sale. JavaScript is used to handle dynamic content, form validation, and user interactions, ensuring a smooth user experience.
- **Backend**
  The backend is built using Node.js, which acts as the server-side environment for handling application logic. It processes user requests, manages APIs, and ensures proper communication between the frontend and the database. The backend is responsible for tasks such as user authentication handling, product data processing, and request-response management. It ensures that all operations are performed securely and efficiently.
- **Database**
  The database is managed using Firebase, which is a cloud-based Backend-as-a-Service (BaaS) platform. Firebase provides real-time data storage, allowing instant updates and synchronization of user and product data. It also offers built-in authentication services for secure login and registration, and cloud storage for storing images of EV parts. This ensures scalable, reliable, and efficient data management for the system.

4.2 Project working with Snapshots

The working of the system can be understood through different modules:

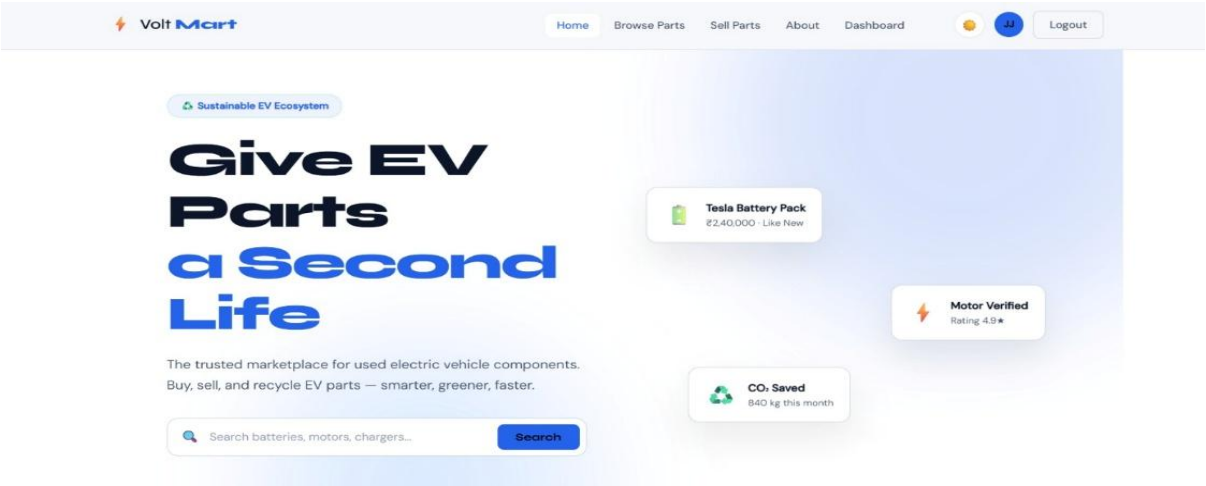


Fig 4.2.1 Home Page of Voltmart

The above figure illustrates the home page of the Voltmart platform, which serves as the primary interface for users accessing the system. This page is designed to provide an overview of the platform's purpose, features, and navigation options in a visually appealing and user-friendly manner. The main objective of the home page is to introduce users to the concept of buying and selling used electric vehicle parts while promoting sustainability and recycling.

At the top of the page, a navigation bar is provided which includes options such as Home, Browse Parts, Sell Parts, About, and Dashboard. This navigation bar allows users to easily move between different sections of the platform without confusion. It also includes user-related options such as profile access and logout functionality, ensuring a personalized experience.

The central section of the page highlights the main slogan of the platform, emphasizing the idea of giving electric vehicle parts a second life. This reflects the core goal of the system, which is to reduce waste and encourage reuse. Supporting elements such as product highlights (e.g., battery pack, verified motor, and environmental impact statistics) are displayed to build user trust and engagement.

A search bar is prominently placed to allow users to quickly search for specific EV parts such as batteries, motors, or chargers. This improves usability by reducing the time required to locate desired items. Overall, the home page is designed to provide an intuitive and informative experience, encouraging users to explore and interact with the platform.

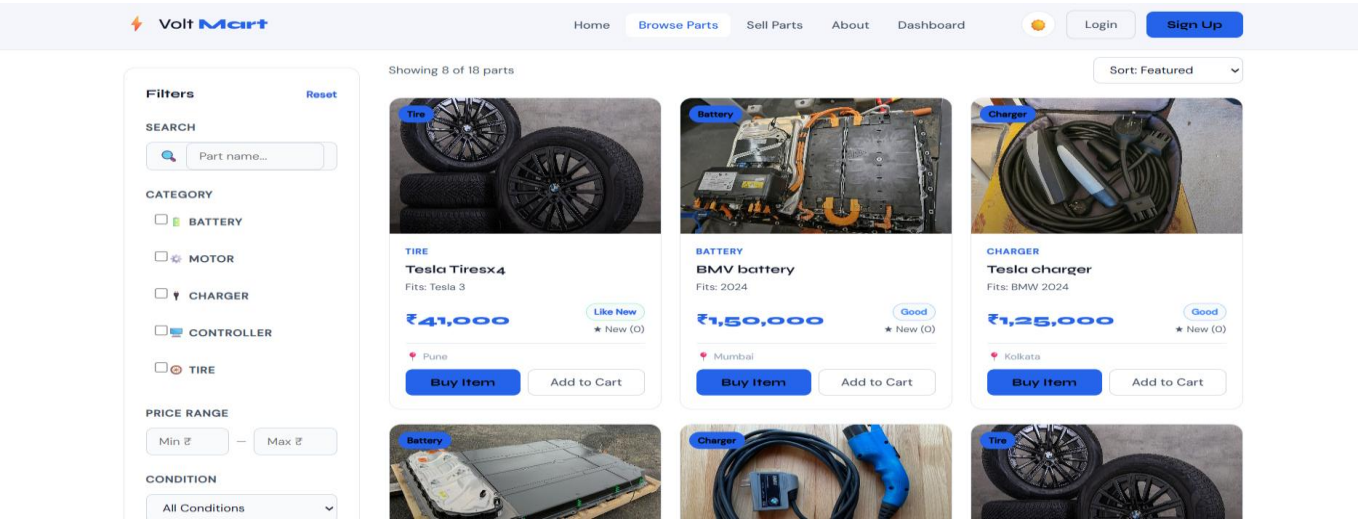


Fig 4.2.2 Browse EV Parts Page

This page represents the **"Browse Parts" marketplace interface** of the Voltmart system, where users can explore and purchase available EV components.

At the top, there is a **navigation bar** that includes options such as Home, Browse Parts, Sell Parts, About, and Dashboard, along with Login and Sign Up buttons. This allows users to easily navigate through different sections of the platform.

On the left side, a **filter panel** is provided to refine search results. Users can search for specific parts using a search bar, filter items by categories like Battery, Motor, Charger, Controller, and Tire, and set a price range or condition. This helps users quickly find relevant products based on their requirements.
The main section displays a **grid of product cards**, each showing details of EV parts such as images, product name, compatibility (e.g., Tesla or BMW), price, condition (like "Good" or "Like New"), and location. Each product card also includes action buttons like **"Buy Item"** and **"Add to Cart"**, allowing users to proceed with purchase or save items.At the top of the product listing, users can see the total number of items available and a **sorting option** (e.g., Featured), which helps organize products based on preferences.
Overall, this page provides a clean, user-friendly interface for browsing, filtering, and purchasing used EV parts efficiently.

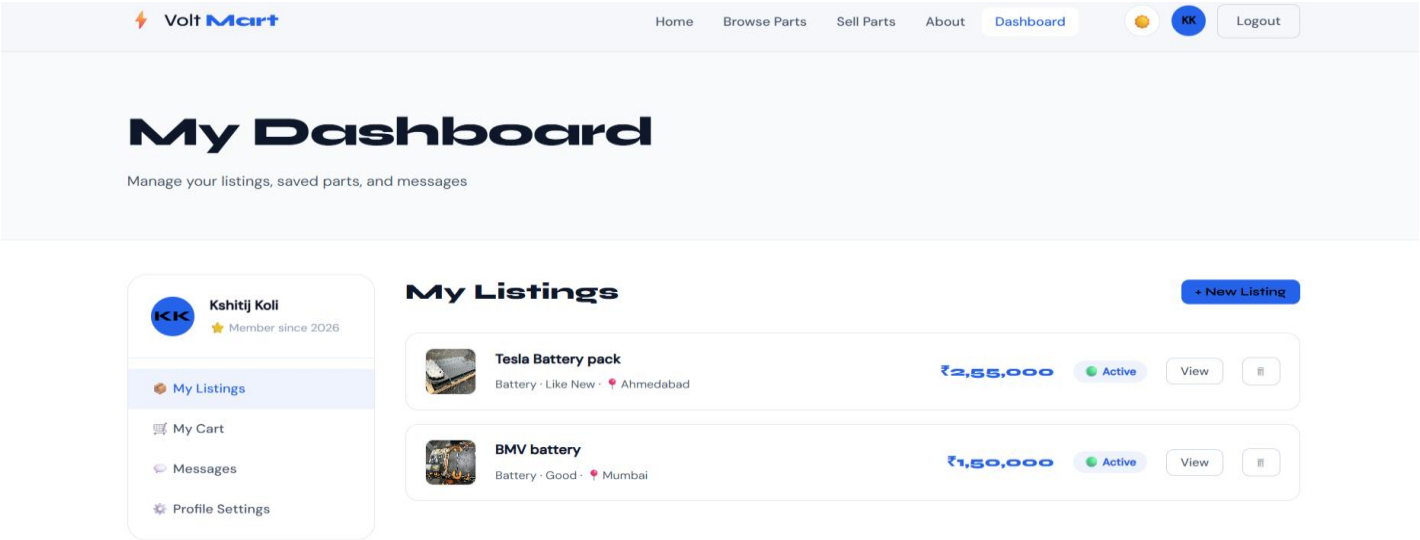
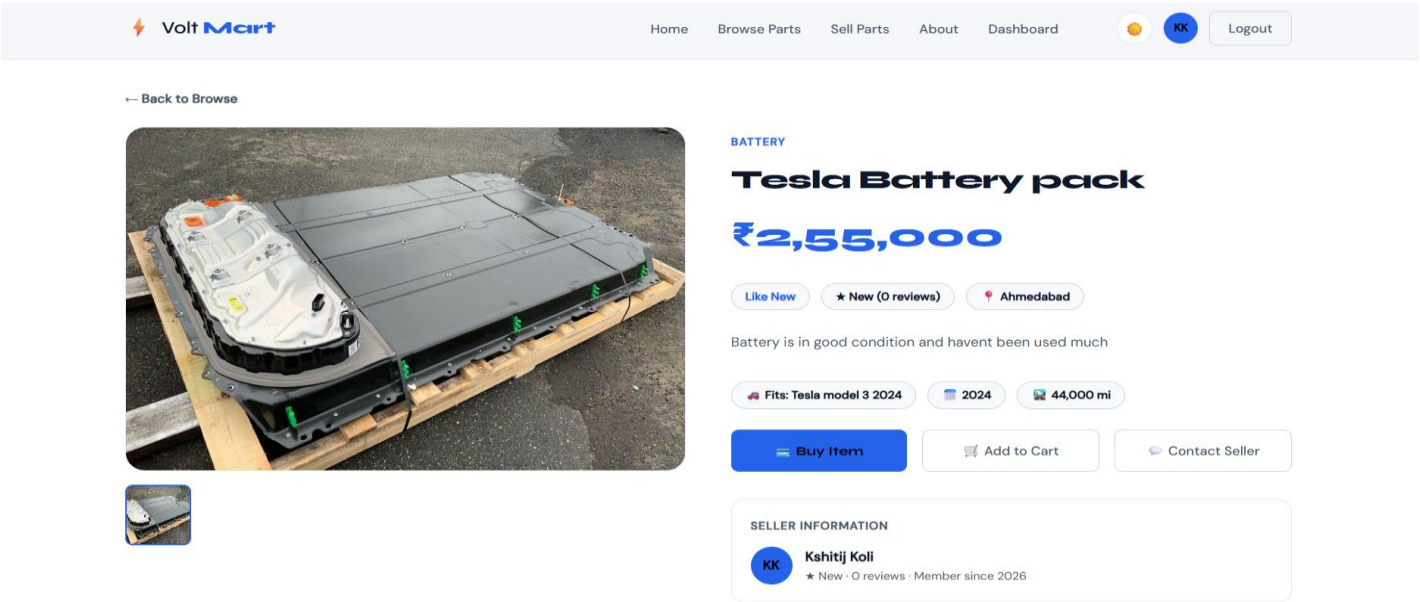


Fig 4.2.3 User Dashboard

This page shows the **Dashboard** section of the Voltmart system, where users can manage their activity.
At the top, there is a navigation bar with options like Home, Browse Parts, Sell Parts, About, and Dashboard, along with a logout option and user profile icon. The main heading **"My Dashboard"** indicates a personalized area for the user.
On the left side, a **sidebar panel** displays the user's profile (name and membership info) and provides quick access to sections like **My Listings, My Cart, Messages, and Profile Settings**.
The main section shows **"My Listings"**, where the user can view all the EV parts they have posted. Each listing includes product image, name, category, condition, location, and price. It also shows the status (e.g., Active) and provides actions like **View** or delete/edit options. A **"+ New Listing"** button allows users to add new products.
It provides a centralized space for users to monitor and control their selling activity efficiently. The dashboard also enhances user experience by offering quick navigation and real-time updates of their listings and account details.

Overall, this page helps users easily manage their listings, track their items, and control their account activities in one place.



4.2.4 EV Part Card

This page shows the **product details view** of a specific EV part in the Voltmart system.At the top, there is a navigation bar and a **"Back to Browse"** option for easy navigation. The left side displays a **large product image** with a preview thumbnail. On the right side, detailed information about the item is shown, including the product name (**Tesla Battery Pack**), price, condition (Like New), location, and a short description.
Additional details such as compatibility (Tesla Model 3 2024), year, and usage (mileage) are also provided. Below this, users have action buttons like **Buy Item**, **Add to Cart**, and **Contact Seller**. At the bottom, **seller information** is displayed, including the seller's name, profile, and membership details.
Overall, this page provides complete information about a product and allows users to take actions like purchasing or contacting the seller.

Chapter 5

Applications and Future Scope

There are various applications of this domain system. The application is listed here.

5.1 Applications

- **EV Parts Marketplace:** Voltmart provides a dedicated online platform where users can buy and sell used electric vehicle parts in an organized and efficient way.
- **Cost Reduction for Users:** It helps EV owners reduce maintenance expenses by offering affordable second-hand parts instead of costly new components.
- **Support for Repair & Service Centers:** Mechanics and EV service providers can quickly find required parts, saving time and improving service efficiency.
- **Promotes Recycling & Sustainability:** The platform encourages reuse of functional EV components, reducing electronic waste and supporting eco-friendly practices.
- **Wider Market Access:** Sellers, including small businesses and scrap dealers, can reach a larger audience beyond their local area.

5.2 Future Scope

- **Mobile Application Development:** A dedicated mobile app can be developed to improve user accessibility and provide a smoother experience on smartphones.
- **AI-Based Recommendations:** Smart algorithms can suggest relevant parts to users based on their search history and vehicle type.
- **Logistics & Delivery Integration:** Partnering with delivery services can enable doorstep shipping of EV parts across regions.
- **User Ratings & Reviews:** Introducing feedback systems will help build trust and improve the quality of sellers and products.
- **Global Expansion:** The platform can be scaled internationally, allowing users from different countries to trade EV parts.
- **Enhanced Security with Blockchain:** Blockchain technology can be used in the future to ensure transparent and secure transactions.

Chapter 6

Summary

The **Voltmart system** is a web-based application developed to provide a dedicated platform for buying and selling used electric vehicle (EV) parts. With the increasing adoption of EVs, the demand for spare parts has grown significantly, but the high cost of new components creates a challenge for users. At the same time, many usable EV parts are discarded due to lack of a proper resale system. Voltmart addresses this gap by offering a centralized and organized marketplace where users can easily trade used EV components.

The platform allows users to register, log in securely, list their products, browse available parts, and purchase items based on their needs. It includes features such as search and filtering options, product listings, and user dashboards, making the system efficient and easy to use. The interface is designed to be interactive and responsive, ensuring a smooth user experience.

The system is developed using **HyperText Markup Language (HTML)**, **Cascading Style Sheets (CSS)**, and **JavaScript (JS)** for the frontend, while **Node.js** is used for backend development to handle application logic and server-side operations. **Firebase**, a cloud-based Backend-as-a-Service platform, is used for real-time database management, user authentication, and cloud storage.

Overall, Voltmart helps reduce the cost of EV maintenance, promotes reuse and recycling of components, and minimizes electronic waste. It provides a scalable, secure, and efficient solution that benefits both buyers and sellers while contributing to environmental sustainability.

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